

A Unified Framework and Sparse Bayesian Perspective for Direction-of-Arrival Estimation in the Presence of Array Imperfections

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Abstract—Self-calibration methods play an important role in reducing the negative effects of array imperfections during direction-of-arrival (DOA) estimation. However, the dependence of most such methods on the eigenstructure techniques greatly degrades their adaptation to demanding scenarios, such as low signal-to-noise ratio (SNR) and limited snapshots. This paper aims at formulating a unified framework and sparse Bayesian perspective for array calibration and DOA estimation. A comprehensive model of the array output is presented first when a single type of array imperfection is considered, with mutual coupling, gain/phase uncertainty, and sensor location error treated as typical examples. The spatial sparsity of the incident signals is then exploited, and a Bayesian method is proposed to realize array calibration and source DOA estimation. The array perturbation magnitudes are assumed to be small according to most application scenarios, and the geometries of mutually coupled arrays are assumed to be uniform linear and those of arrays with sensor location errors are assumed to be linear. Cramer–Rao lower bounds (CRLBs) for the array calibration and DOA estimation precisions are also obtained. The sparse Bayesian method is finally extended to deal with the DOA estimation problem when more than one type of array perturbation coexists.

Index Terms—Direction-of-arrival (DOA) estimation, array calibration, perturbed array output formulation, Cramer–Rao lower bound (CRLB), sparse Bayesian reconstruction.

I. INTRODUCTION

MOST of the existing high-resolution direction-of-arrival (DOA) estimation methods formulate the array output as a parametric function of the signal directions, and their performance may deteriorate significantly in the presence of array imperfections. Among those imperfections, the mutual coupling, gain/phase uncertainty and sensor location error are very typical ones, and they can hardly be calibrated completely via antenna design.

The experimental methods for array calibration and DOA estimation are generally very expensive, and the calibration result should be updated once the environment changes. Moreover, experimental calibration does not work for the satellite- and air-borne array systems [1]. Therefore, the self-calibration and

DOA estimation methods have aroused much research interest during the past few years. Those methods formulate the array perturbations with determined but unknown parameters, and estimate those parameters together with the source directions, so as to realize array calibration and DOA estimation. Most of the existing methods focus on a certain type of array imperfection, such as mutual coupling [2]–[4], gain/phase uncertainty [5]–[7] and sensor location error [8], [9]. Those methods are generally eigenstructure-based, and they may lack adaptation to demanding scenarios with low signal-to-noise ratio (SNR), limited snapshots and spatially adjacent sources, just as their counterparts do in accurately calibrated arrays. The DOA estimation problem in the presence of more than one type of array perturbation has also been studied in a small amount of literatures [10], [11]. Nonetheless, the method in [10] is computationally much expensive as it requires multi-dimensional parameter searching, while the method in [11] needs to know the directions of the calibration sources.

Theoretical analysis of the array calibration and DOA estimation precisions has also been a hot topic in this area, and the corresponding Cramer–Rao lower bounds (CRLB) are available in [12]–[15], but those results adapt to certain array perturbation types only.

The recently interest-attracting sparse reconstruction techniques have also been introduced into the array signal processing area. Related methods make use of the spatial sparsity of the incident signals, and have been shown to gain much enhanced adaptation over the eigenstructure-based methods to low SNR, limited snapshots and spatially adjacent signals [16]–[30]. The existing sparsity-promoting DOA estimation methods based on accurately calibrated arrays generally divide into two categories, the ℓ_p -norm ($0 \leq p \leq 1$) based ones [16]–[28] and the sparse Bayesian learning (SBL) based ones [29], [30]. As the SBL technique is able to reduce the structural and convergence errors in total when compared with the ℓ_p -norm based technique during sparse reconstruction [31], [32], the SBL-based DOA estimators have been demonstrated empirically to achieve higher precision than the ℓ_p -norm based counterparts [30]. However, the spatial reconstruction models of the methods in both categories will become inaccurate if array perturbations are present but overlooked, and how to remove the negative effects of those perturbations during the sparsity-promoting DOA estimation process still requires further research.

This paper follows a unified framework to address the DOA estimation problem in the presence of array imperfections, with *typical perturbations* of mutual coupling, gain/phase uncertainty and sensor location error taken into consideration.

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Assumptions of small magnitudes are added for the perturbations as it is the fact in most practical applications, and linear and uniform linear array geometries are assumed for sensor location error and mutual coupling effect, respectively, to facilitate notation and method derivation. Based on the assumptions, a comprehensive perturbed array output model is established first, and a SBL-based method named Sparse Bayesian Array Calibration (SBAC) is then proposed to realize array calibration and DOA estimation by exploiting the spatial sparsity of the incident signals. After that, the CRLB of the array calibration and DOA estimation precisions are derived based on the comprehensive array output model, the results apply to the array signal processing scenarios in the presence of any type of the typical array imperfections. Finally, the scenario when mutual coupling and gain/phase uncertainty coexist is taken as an example, so as to show how the SBAC method can be extended to calibrate more than one type of array perturbations and realize DOA estimation simultaneously. Numerical examples will also be provided to show how the proposed method performs in array calibration and DOA estimation.

Notations: $(\bullet)^*$, $(\bullet)^T$ and $(\bullet)^H$ denote conjugate, transpose and conjugate transpose, respectively, $\text{toeplitz}(\bullet)$ forms a Toeplitz matrix by taking the given vector as the first column, bold lower- and upper-case variables stand for vectors and matrices, respectively, z_p represents the p th element of $\mathbf{z}$, $\mathbf{Z}_{p,:}$, $\mathbf{Z}_{:,p}$ and $\mathbf{Z}_{p_1,p_2}$ represent the p th row, p th column and (p_1, p_2) th element of $\mathbf{Z}$, respectively, $\|\bullet\|_\kappa$ denotes the κ -norm of a vector and $\|\bullet\|_F$ denotes the Frobenius norm of a matrix, $\text{diag}(\bullet)$ means forming a diagonal matrix by taking the given vector as the main diagonal, $\text{Re}(\bullet)$ and $\text{Im}(\bullet)$ mean taking the real and imaginary parts of a complex value, respectively, $\text{tr}\{\bullet\}$ is the trace operator, $\mathbb{E}\{\bullet\}$ is the expectation operator, $\odot$ stands for the element-wise product of two matrices, $\emptyset$ represents the null set, $\mathbf{I}_M$ stands for the identity matrix with dimension $M \times M$, $j = \sqrt{-1}$.

II. PERTURBED ARRAY OUTPUT FORMULATION

Suppose that K narrowband signals impinge onto an M -element array from directions of $\boldsymbol{\vartheta} = [\vartheta_1, \dots, \vartheta_K]^T$, N snapshots are collected, and the output of the accurately calibrated array at time t is

$$\mathbf{x}^{(pf)}(t) = \sum_{k=1}^K \mathbf{a}(\vartheta_k) s_k(t) + \mathbf{v}(t) = \mathbf{A}(\boldsymbol{\vartheta}) \mathbf{s}(t) + \mathbf{v}(t), \quad (1)$$

where $\mathbf{A}(\boldsymbol{\vartheta}) = [\mathbf{a}(\vartheta_1), \dots, \mathbf{a}(\vartheta_K)]$ and $\mathbf{a}(\vartheta_k) = [e^{j\varphi_{k,1}}, \dots, e^{j\varphi_{k,M}}]^T$ are the array responding matrix and vector, respectively, $\varphi_{k,m}$ is the phase-delay of the k th signal when propagating from the reference to the m th sensor, $s_k(t)$ is the waveform of the k th signal at time t , $\mathbf{s}(t) = [s_1(t), \dots, s_K(t)]^T$, $\mathbf{v}(t) = [v_1(t), \dots, v_M(t)]^T$ is the noise vector with variance σ^2 ; $\mathbf{x}^{(pf)}(t) = [x_1^{(pf)}(t), \dots, x_M^{(pf)}(t)]^T$ and the superscript (pf) is the short of “perturbation-free”.

When array imperfections are taken into consideration, the array output is given as follows,

$$\mathbf{x}(t) = \sum_{k=1}^K \mathbf{a}'(\vartheta_k) s_k(t) + \mathbf{v}(t) = \mathbf{A}'(\boldsymbol{\vartheta}) \mathbf{s}(t) + \mathbf{v}(t), \quad (2)$$

where $\mathbf{A}'(\boldsymbol{\vartheta}) = [\mathbf{a}'(\vartheta_1), \dots, \mathbf{a}'(\vartheta_K)]$, and $\mathbf{a}'(\vartheta)$ is the perturbed responding vector. The three types of typical array perturbations, including mutual coupling, gain/phase uncertainty and sensor location error, are concerned in this paper, and the array geometry is assumed to be linear to simplify notations. The sensor location error is also assumed to exist along the array axes and thus does not destroy the linear geometry of the array. The analysis of the mutual coupling effect is further restricted to uniform linear arrays (ULA). The responding vector of the accurately calibrated linear array is $\mathbf{a}(\vartheta_k) = [e^{j2\pi d_1 \sin \vartheta_k / \lambda}, \dots, e^{j2\pi d_M \sin \vartheta_k / \lambda}]^T$, i.e., $\varphi_{k,m} = 2\pi d_m \sin \vartheta_k / \lambda$ in (1), where λ represents the signal wavelength and $d_1, \dots, d_M$ are the distances of the M array sensors from the reference located on the array axes.

The perturbed array responding vector has diverse expressions in the case of different array imperfections. When mutual coupling or gain/phase uncertainty is present, the vector is $\mathbf{a}'(\vartheta) = \mathbf{C}\mathbf{a}(\vartheta)$ [2]–[7], with $\mathbf{C}$ representing the mutual coupling or gain-phase matrix, respectively. The mutual coupling matrix can be written more explicitly as $\mathbf{C} = \text{toeplitz}([1, b_1, \dots, b_P, \mathbf{0}_{(M-P-1) \times 1}^T])$, where $b_p \in \mathbb{C}$ is the coupling coefficient of the two sensors displaced by $p - 1$ times the inter-element spacing of the ULA, and b_p is very small when $p > P$ and is thus neglected. The gain-phase matrix is $\mathbf{C} = \text{diag}([\alpha_1 e^{j\phi_1}, \dots, \alpha_M e^{j\phi_M}])^T$, where $\alpha_1, \dots, \alpha_M$ and $\phi_1, \dots, \phi_M$ are the gains and initial phases of the sensors. The array responding vector in the presence of sensor location error is $\mathbf{a}'(\vartheta) = [e^{j2\pi(d_1 + \tilde{d}_1) \sin \vartheta / \lambda}, \dots, e^{j2\pi(d_M + \tilde{d}_M) \sin \vartheta / \lambda}]^T$, with $\tilde{d}_1, \dots, \tilde{d}_M$ being the location errors of the M sensors [8], [9]. In this paper, we take the first array sensor as the reference, thus $\alpha_1 = 1$, $\phi_1 = 0$, $d_1 = 0$ and $\tilde{d}_1 = 0$.

Although the directional information of the incident signals is still reserved in the perturbed array outputs, it can hardly be extracted directly based on the array output formulation in (2), as the structure of $\mathbf{A}'(\boldsymbol{\vartheta})$ is not available beforehand due to the unknown array imperfections. In order to highlight the perturbation-free signal components, and also make clearer the relationship between the array output and the perturbation parameters, we propose to establish the following comprehensive formulation of the perturbed array output,

$$\mathbf{x}(t) = \mathbf{A}'(\boldsymbol{\vartheta}) \mathbf{s}(t) + \mathbf{v}(t) = \mathbf{A}(\boldsymbol{\vartheta}) \mathbf{s}(t) + \mathbf{Q}_t \mathbf{c} + \mathbf{v}(t). \quad (3)$$

(3) adapts to any of the typical array perturbations, including mutual coupling, gain/phase uncertainty and sensor location error, with $\mathbf{c}$ standing for a column vector consisting of the array perturbation parameters, $\mathbf{c} \in \mathbb{C}^{P \times 1}$ for mutual coupling, $\mathbf{c} \in \mathbb{C}^{(M-1) \times 1}$ for gain/phase uncertainty and $\mathbf{c} \in \mathbb{R}^{(M-1) \times 1}$ for sensor location error. We use P to denote the dimension of $\mathbf{c}$ uniformly for notational convenience, i.e., $\mathbf{c} = [c_1, \dots, c_P]^T$, and $P = M - 1$ in the cases of gain/phase uncertainty and sensor location error. The explicit formulation of $\mathbf{A}'(\boldsymbol{\vartheta})$ varies with respect to (w.r.t.) the array perturbation type, and the following equation holds for $\mathbf{Q}_t$ and $\mathbf{c}$,

$$\mathbf{Q}_t \mathbf{c} = [\mathbf{A}'(\boldsymbol{\vartheta}) - \mathbf{A}(\boldsymbol{\vartheta})] \mathbf{s}(t) \triangleq \boldsymbol{\Psi} \boldsymbol{\Phi}(\boldsymbol{\vartheta}) \mathbf{s}(t), \quad (4)$$

where $\boldsymbol{\Psi}$ is a function of $\mathbf{c}$ and independent of the signal directions, while $\boldsymbol{\Phi}(\boldsymbol{\vartheta})$ is independent of $\mathbf{c}$ and relies on the signal directions, they diverge largely according to the array perturbation type. However, for each of the typical perturbations, it holds

for $\mathbf{Q}_t$ that $[\mathbf{Q}_t]_{:,p} = \mathbf{G}_p \Phi(\boldsymbol{\vartheta}) \mathbf{s}(t)$ with $\mathbf{G}_p = \partial \Psi / \partial c_p$. Such an expression of $\mathbf{Q}_t$ is concluded by taking the differentiations of both sides of (3) w.r.t. c_p .

In order to explain (3) and (4) more explicitly, we itemize the expressions of $\mathbf{c}$, Ψ , $\Phi(\boldsymbol{\vartheta})$ and $\mathbf{Q}_t$ corresponding to different array perturbations in the following. For notational convenience, the three kinds of array perturbations will be classified as Type I (mutual coupling), Type II (gain/phase uncertainty) and Type III (sensor location error), respectively, in the rest of the paper, and the superscripts of (I), (II) and (III) will be used when necessary to clarify the particular perturbation type.

Mutual Coupling: Equation (2) can be rewritten as follows in the presence of mutual coupling,

$$\begin{aligned} \mathbf{x}(t) &= \mathbf{A}(\boldsymbol{\vartheta}) \mathbf{s}(t) + \left(\mathbf{C}^{(I)} - \mathbf{I}_M \right) \mathbf{A}(\boldsymbol{\vartheta}) \mathbf{s}(t) + \mathbf{v}(t) \\ &= \mathbf{A}(\boldsymbol{\vartheta}) \mathbf{s}(t) + \mathbf{Q}_t^{(I)} \mathbf{c}^{(I)} + \mathbf{v}(t), \end{aligned} \quad (5)$$

By combining (5) and (4), it can be concluded that, $\mathbf{c}^{(I)} = [b_1, \dots, b_P]^T$, $\Psi^{(I)} = \text{toeplitz}([0, (\mathbf{c}^{(I)})^T, \mathbf{0}_{(M-P-1) \times 1}^T])^T$, $\Phi^{(I)}(\boldsymbol{\vartheta}) = \mathbf{A}(\boldsymbol{\vartheta})$, $\mathbf{Q}_t^{(I)} \in \mathcal{C}^{M \times P}$, $[\mathbf{Q}_t^{(I)}]_{:,p} = \mathbf{G}_p^{(I)} \mathbf{A}(\boldsymbol{\vartheta}) \mathbf{s}(t)$ and $\mathbf{G}_p^{(I)} = \partial \mathbf{C}^{(I)} / \partial c_p^{(I)} \in \mathcal{R}^{M \times M}$ contains nonzero elements of 1 only on the $\pm p$ diagonals.

Gain/Phase Uncertainty: Equation (2) can also be rewritten in a similar form as that given in (5) in the presence of gain/phase uncertainty, with $\Phi^{(II)}(\boldsymbol{\vartheta}) = \mathbf{A}(\boldsymbol{\vartheta})$ and some differences lying in $\mathbf{c}^{(II)} = [c_1^{(II)}, \dots, c_{M-1}^{(II)}]^T$, $c_p^{(II)} = \alpha_{p+1} e^{j\phi_{p+1}} - 1$, $\Psi^{(II)} = \text{diag}([0, (\mathbf{c}^{(II)})^T])^T$, $\mathbf{Q}_t^{(II)} \in \mathcal{C}^{M \times (M-1)}$, $[\mathbf{Q}_t^{(II)}]_{:,p} = \mathbf{G}_p^{(II)} \mathbf{A}(\boldsymbol{\vartheta}) \mathbf{s}(t)$ and $\mathbf{G}_p^{(II)} = \partial \mathbf{C}^{(II)} / \partial c_p^{(II)} \in \mathcal{R}^{M \times M}$ has its $(p+1, p+1)$ th element being the only nonzero value of 1.

Sensor Location Error: The array output formulation in the presence of sensor location error is

$$\begin{aligned} \mathbf{x}(t) &= \mathbf{A}(\boldsymbol{\vartheta}) \mathbf{s}(t) + \text{diag} \left([0, \tilde{d}_2, \dots, \tilde{d}_M]^T \right) \\ &\quad \times \mathbf{A}(\boldsymbol{\vartheta}) \text{diag} \left(\frac{j2\pi}{\lambda} [\sin \vartheta_1, \dots, \sin \vartheta_K]^T \right) \\ &\quad \times \mathbf{s}(t) + \mathbf{v}(t), \\ &= \mathbf{A}(\boldsymbol{\vartheta}) \mathbf{s}(t) + \mathbf{Q}_t^{(III)} \mathbf{c}^{(III)} + \mathbf{v}(t). \end{aligned} \quad (6)$$

The first equation in (6) is concluded via first-order Taylor expansion under the assumption of small sensor location errors [8], and the corresponding parameters are defined as $\mathbf{c}^{(III)} = [c_1^{(III)}, \dots, c_{M-1}^{(III)}]^T$, $c_p^{(III)} = \tilde{d}_{p+1}$, $\Psi^{(III)} = \text{diag}([0, (\mathbf{c}^{(III)})^T])^T$, $\Phi^{(III)}(\boldsymbol{\vartheta}) = \mathbf{A}(\boldsymbol{\vartheta}) \text{diag}(\frac{j2\pi}{\lambda} [\sin \vartheta_1, \dots, \sin \vartheta_K]^T)$, $\mathbf{Q}_t^{(III)} \in \mathcal{C}^{M \times (M-1)}$, $[\mathbf{Q}_t^{(III)}]_{:,p} = \mathbf{G}_p^{(III)} \Phi^{(III)}(\boldsymbol{\vartheta}) \mathbf{s}(t)$ and $\mathbf{G}_p^{(III)} = \mathbf{G}_p^{(II)}$.

The perturbed array output formulation given in (3) reveals the relationship between the measurement vectors and the perturbation parameters, and such a relationship can be exploited to estimate the parameters by minimizing the fitting error between $\mathbf{A}(\boldsymbol{\vartheta}) \mathbf{s}(t) + \mathbf{Q}_t \mathbf{c}$ and $\mathbf{x}(t)$. Meanwhile, for a particular perturbation estimate, the fitting error between $\mathbf{A}'(\boldsymbol{\vartheta}) \mathbf{s}(t)$ and $\mathbf{x}(t)$ can be minimized to determine the signal directions. In the next section, we follow this guideline to calibrate the array imperfections and estimate the signal directions iteratively. The spatial sparsity of the incident signals will also be taken into consideration to enhance the adaptation of the method to demanding

scenarios, just as the sparsity-promoting DOA estimators have done in the accurately calibrated arrays [16]–[30].

III. SPARSE BAYESIAN ARRAY CALIBRATION AND DOA ESTIMATION

In this part, we propose a sparsity-promoting method to re-align array calibration and DOA estimation based on the perturbed array output formulation given in (3).

A. Spatially Overcomplete Array Output Formulation

In order to highlight the spatial sparsity of the incident signals, the array output should be reformulated in an overcomplete form. The potential space of the incident signals can be sampled discretely to form a direction set of $\Theta = [\theta_1, \dots, \theta_L]$, and the true source directions $\vartheta_1, \dots, \vartheta_K$ are contained in Θ with moderately small quantization errors. The variables that depend on $\boldsymbol{\vartheta}$ in Section II can be zero-padded according to Θ to form their overcomplete variations, which are identified with (Θ) , e.g., $\mathbf{A}(\Theta) = [\mathbf{a}(\theta)]_{\theta \in \Theta}$, then (3) can be extended as follows,

$$\mathbf{x}(t) = \mathbf{A}'(\Theta) \bar{\mathbf{s}}(t) + \mathbf{v}(t) = \mathbf{A}(\Theta) \bar{\mathbf{s}}(t) + \bar{\mathbf{Q}}_t \mathbf{c} + \mathbf{v}(t), \quad (7)$$

where $\bar{\mathbf{s}}(t)$ is the zero-padded variation of $\mathbf{s}(t)$ from $\boldsymbol{\vartheta}$ to Θ , $\bar{\mathbf{Q}}_t$ and $\mathbf{c}$ satisfy $\bar{\mathbf{Q}}_t \mathbf{c} = [\mathbf{A}'(\Theta) - \mathbf{A}(\Theta)] \bar{\mathbf{s}}(t) = \Psi \Phi(\Theta) \bar{\mathbf{s}}(t)$, the p th column of $\bar{\mathbf{Q}}_t \in \mathcal{C}^{M \times P}$ is $[\bar{\mathbf{Q}}_t]_{:,p} = \mathbf{G}_p \Phi(\Theta) \bar{\mathbf{s}}(t)$ with $\mathbf{G}_p = \partial \Psi / \partial c_p$. As the true source directions $\boldsymbol{\vartheta}$ are not included in Θ with probability 1, (7) is not an accurate extension of (3). However, previous results in this field have shown that, the inaccurate overcomplete model works satisfyingly given that the quantization error of Θ is adequately small, and the spatial signal distribution can be reconstructed from the measurements based on the overcomplete model effectively [16]–[30], [33]. Therefore, the model quantization error in (7) is overlooked in the rest of the paper unless otherwise stated.

By extending the standard perturbed array output formulation in (3) to the spatially overcomplete one in (7), the original DOA estimation problem changes to one of approximating the measurement vectors with candidate signal bases $\mathbf{A}(\Theta) \bar{\mathbf{s}}(t)$ and the corresponding perturbation component $\bar{\mathbf{Q}}_t \mathbf{c}$. The approximation criterion should be defined based on the fact that, only a small number of signals may impinge simultaneously, which results in a sparsity constraint for the overcomplete model. Effective exploitation of such a sparsity constraint has been demonstrated empirically to enhance the superresolution ability of DOA estimators largely, especially in much demanding scenarios with low SNR, limited snapshots and spatially adjacent sources [16]–[30]. Robust subspace estimation in such scenarios has also been a hot topic in the signal processing community. O. Besson et. al. resorted to the Bayesian technique and contributed a lot to the solving of this problem [34], [35]. Their work improves the subspace estimation precision generally, and can also be taken as a reference for DOA estimation. However, as the prior assumption about the subspace is added by taking the (multi-dimensional) subspace as a whole, the major concern in their work is also the total estimation precision of the subspace. Therefore, in order to introduce their ideas into the array signal processing area, further research is still required to decouple the subspaces and obtain refined DOA estimates.

B. Spatial Signal Reconstruction and Perturbation Parameter Estimation

A key point for reconstructing the signal components based on (7) is adding the sparsity constraint properly to the overcomplete model, so as to eliminate the irrelevant bases. Meanwhile, the coupling between different signal components should be avoided to reconstruct each of them as accurately as possible. In this subsection, we introduce the SBL technique and use a group of independent parameters to represent the signal variances in different directions, and optimize the parameters iteratively based on the array outputs. When most of the parameters converge to 0, the number and indexes of the surviving parameters indicate the number and locations of the incident signals.

The hyperparameter set $\gamma = [\gamma_1, \dots, \gamma_L]^T$ is introduced to represent the variances of the zero-mean signal waveforms in the directions of $\theta_1, \dots, \theta_L$, i.e.,

$$\bar{\mathbf{s}}(t) \sim \mathcal{N}(\mathbf{0}, \mathbf{\Gamma}) \quad (8)$$

where $\mathbf{\Gamma} = \text{diag}(\gamma)$. As $\bar{\mathbf{s}}(t)$ is the zero-padded extension of $\mathbf{s}(t)$ from $\boldsymbol{\vartheta}$ to $\boldsymbol{\Theta}$, γ should only contain K nonzero elements associated with $\boldsymbol{\vartheta}$, and it is highly sparse as $K \ll L$. In the above distribution function, a single parameter is used to describe the time-varying waveform of the signal located in the corresponding direction, thus the spatial signal distribution can be reconstructed more easily by estimating the $L \times 1$ vector γ , instead of the $L \times N$ matrix $[\bar{\mathbf{s}}(1), \dots, \bar{\mathbf{s}}(N)]$. Since only particular values are concerned for $\mathbf{c}$ and σ^2 , indicative functions are introduced to describe their prior distributions. Based on the distribution assumptions of the parameters, we derive the statistical dependence of the array outputs on the hyperparameters in this subsection, so as to look for maximum likelihood estimates of the parameters and realize array calibration and spatial signal reconstruction.

The likelihood function of the perturbed array output derived from (7) is

$$\begin{aligned} p(\{\mathbf{x}(t)\}_{t=1}^N | \{\bar{\mathbf{s}}(t)\}_{t=1}^N; \mathbf{c}, \sigma^2) \\ = |\pi \sigma^2 \mathbf{I}_M|^{-N} \exp \left\{ -\sigma^{-2} \sum_{t=1}^N \|\mathbf{x}(t) - \mathbf{A}'(\boldsymbol{\Theta})\bar{\mathbf{s}}(t)\|_2^2 \right\} \\ = |\pi \sigma^2 \mathbf{I}_M|^{-N} \exp \left\{ -\sigma^{-2} \sum_{t=1}^N \|\mathbf{x}(t) - \mathbf{A}(\boldsymbol{\Theta})\bar{\mathbf{s}}(t) - \bar{\mathbf{Q}}_t \mathbf{c}\|_2^2 \right\}. \end{aligned} \quad (9)$$

Combining (8) and (9) yields the following probability density of $\{\mathbf{x}(t)\}_{t=1}^N$ w.r.t. the hyperparameters,

$$\begin{aligned} p(\{\mathbf{x}(t)\}_{t=1}^N; \gamma, \mathbf{c}, \sigma^2) \\ = \int \cdots \int p(\{\mathbf{x}(t)\}_{t=1}^N | \{\bar{\mathbf{s}}(t)\}_{t=1}^N; \mathbf{c}, \sigma^2) \\ \times p(\{\bar{\mathbf{s}}(t)\}_{t=1}^N; \gamma) d\bar{\mathbf{s}}(1) \cdots d\bar{\mathbf{s}}(N) \\ = \prod_{t=1}^N \int p(\mathbf{x}(t) | \bar{\mathbf{s}}(t); \mathbf{c}, \sigma^2) p(\bar{\mathbf{s}}(t); \gamma) d\bar{\mathbf{s}}(t) \\ = |\pi \Sigma_{\mathbf{x}}|^{-N} \exp \left\{ -N \text{tr} \left(\Sigma_{\mathbf{x}}^{-1} \hat{\mathbf{R}}_{\mathbf{x}} \right) \right\}, \end{aligned} \quad (10)$$

where $\Sigma_{\mathbf{x}} = \sigma^2 \mathbf{I}_M + \mathbf{A}'(\boldsymbol{\Theta})\mathbf{\Gamma}(\mathbf{A}'(\boldsymbol{\Theta}))^H$ and $\hat{\mathbf{R}}_{\mathbf{x}} = \frac{1}{N} \sum_{t=1}^N \mathbf{x}(t)\mathbf{x}^H(t)$.

The hyperparameters of γ , $\mathbf{c}$ and σ^2 can be estimated by maximizing the density function in (10), and the estimates indicate the spatial signal distribution and the array perturbation error. However, straightforward maximization of the density function is hardly achievable due to its high nonlinearity. Therefore, most of the researchers resort to the Expectation-Maximization(EM) algorithm for an iterative solution to similar problems [36]–[39]. Each iteration of the EM algorithm consists of an E-step and an M-step.

E-step:

The expectation of the complete probability $p(\{\mathbf{x}(t)\}_{t=1}^N, \{\bar{\mathbf{s}}(t)\}_{t=1}^N; \gamma, \mathbf{c}, \sigma^2)$ is computed, i.e.,

$$\begin{aligned} F(\{\mathbf{x}(t)\}_{t=1}^N, \{\bar{\mathbf{s}}(t)\}_{t=1}^N; \gamma, \mathbf{c}, \sigma^2) \\ = \langle \ln p(\{\mathbf{x}(t)\}_{t=1}^N, \{\bar{\mathbf{s}}(t)\}_{t=1}^N; \gamma, \mathbf{c}, \sigma^2) \rangle \\ = \langle \ln p(\{\mathbf{x}(t)\}_{t=1}^N | \{\bar{\mathbf{s}}(t)\}_{t=1}^N; \mathbf{c}, \sigma^2) \\ + \ln p(\{\bar{\mathbf{s}}(t)\}_{t=1}^N; \gamma) \rangle, \end{aligned} \quad (11)$$

where $\langle \bullet \rangle$ means computing the conditional expectations w.r.t. the probability of $p(\{\bar{\mathbf{s}}(t)\}_{t=1}^N | \{\mathbf{x}(t)\}_{t=1}^N; \gamma, \mathbf{c}, \sigma^2)$. Substituting (9) and (8) into (11) yields the following probability expectation,

$$\begin{aligned} F(\{\mathbf{x}(t)\}_{t=1}^N, \{\bar{\mathbf{s}}(t)\}_{t=1}^N; \gamma, \mathbf{c}, \sigma^2) \\ = \left\langle -MN \ln \sigma^2 - \sigma^{-2} \sum_{t=1}^N \|\mathbf{x}(t) - \mathbf{A}'(\boldsymbol{\Theta})\bar{\mathbf{s}}(t)\|_2^2 \right. \\ \left. - \sum_{l=1}^L \left[N \ln \gamma_l + \left(\sum_{t=1}^N |\bar{s}_l(t)|^2 \right) / \gamma_l \right] \right\rangle \\ = \left\langle -MN \ln \sigma^2 - \sigma^{-2} \sum_{t=1}^N \|\mathbf{x}(t) - \mathbf{A}(\boldsymbol{\Theta})\bar{\mathbf{s}}(t) - \bar{\mathbf{Q}}_t \mathbf{c}\|_2^2 \right. \\ \left. - \sum_{l=1}^L \left[N \ln \gamma_l + \left(\sum_{t=1}^N |\bar{s}_l(t)|^2 \right) / \gamma_l \right] \right\rangle. \end{aligned} \quad (12)$$

M-step:

The unknown parameters are updated by maximizing $F(\{\mathbf{x}(t)\}_{t=1}^N, \{\bar{\mathbf{s}}(t)\}_{t=1}^N; \gamma, \mathbf{c}, \sigma^2)$, which can be realized by setting its differentiations to those parameters to zero. First, the partial differentiation of the second expression of F in (12) w.r.t. $\mathbf{c}$ is

$$\begin{aligned} \frac{\partial}{\partial \mathbf{c}} F(\{\mathbf{x}(t)\}_{t=1}^N, \{\bar{\mathbf{s}}(t)\}_{t=1}^N; \gamma, \mathbf{c}, \sigma^2) \\ = -2\sigma^{-2} \left\langle \sum_{t=1}^N \bar{\mathbf{Q}}_t^H \bar{\mathbf{Q}}_t \right\rangle \mathbf{c} \\ - \left\langle \sum_{t=1}^N \bar{\mathbf{Q}}_t^H (\mathbf{x}(t) - \mathbf{A}(\boldsymbol{\Theta})\bar{\mathbf{s}}(t)) \right\rangle. \end{aligned} \quad (13)$$

Then differentiating the first expression of F in (12) w.r.t. σ^2 and γ yields

$$\begin{aligned} \frac{\partial}{\partial \sigma^2} F(\{\mathbf{x}(t)\}_{t=1}^N, \{\bar{\mathbf{s}}(t)\}_{t=1}^N; \gamma, \mathbf{c}, \sigma^2) \\ = -\frac{MN}{\sigma^2} + \frac{1}{(\sigma^2)^2} \left\langle \sum_{t=1}^N \|\mathbf{x}(t) - \mathbf{A}'(\boldsymbol{\Theta})\bar{\mathbf{s}}(t)\|_2^2 \right\rangle, \end{aligned} \quad (14)$$

$$\begin{aligned} & \frac{\partial}{\partial \gamma_l} F(\{\mathbf{x}(t)\}_{t=1}^N, \{\bar{\mathbf{s}}(t)\}_{t=1}^N; \boldsymbol{\gamma}, \mathbf{c}, \sigma^2) \\ &= -\frac{N}{\gamma_l} + \frac{1}{(\gamma_l)^2} \left\langle \sum_{t=1}^N |\bar{s}_l(t)|^2 \right\rangle. \end{aligned} \quad (15)$$

The estimates of the variables in the q th EM iteration can be obtained by setting the differentiations in (13)–(15) to 0 as follows,

$$\mathbf{c}^{(q)} = \left\langle \sum_{t=1}^N \bar{\mathbf{Q}}_t^H \bar{\mathbf{Q}}_t \right\rangle^{-1} \left\langle \sum_{t=1}^N \bar{\mathbf{Q}}_t^H (\mathbf{x}(t) - \mathbf{A}(\boldsymbol{\Theta}) \bar{\mathbf{s}}(t)) \right\rangle, \quad (16)$$

$$(\sigma^2)^{(q)} = \frac{1}{MN} \left\langle \sum_{t=1}^N \left\| \mathbf{x}(t) - (\mathbf{A}'(\boldsymbol{\Theta}))^{(q)} \bar{\mathbf{s}}(t) \right\|_2^2 \right\rangle, \quad (17)$$

$$\gamma_l^{(q)} = \frac{1}{N} \left\langle \sum_{t=1}^N |\bar{s}_l(t)|^2 \right\rangle, \quad (18)$$

where the superscript (q) stands for the iteration index, the structure of $(\mathbf{A}'(\boldsymbol{\Theta}))^{(q)}$ depends on $\mathbf{c}^{(q)}$. As the expectations in (16)–(18) rely on the conditional probability $p(\{\bar{\mathbf{s}}(t)\}_{t=1}^N | \{\mathbf{x}(t)\}_{t=1}^N; \boldsymbol{\gamma}, \mathbf{c}, \sigma^2)$, and the formulation of $\bar{\mathbf{Q}}_t$ is much complicated, the iteration strategy of those parameters given above is not usable directly. In Appendix A, we carry out some deeper analysis to facilitate the implementation of the EM iterations.

The conditional expectations given in (46)–(49) can be substituted into (16)–(18) to update $\mathbf{c}$, σ^2 and $\boldsymbol{\gamma}$, and those updated parameters are passed to the next EM iteration. As the EM algorithm has proved convergence property [40], [41], the parameters approach stationary estimates gradually. When a predefined termination criterion is satisfied, the iteration process can be stopped to export the reconstruction result.

Although the EM iteration process plays an important role in estimating the hyperparameters, the hierarchical probabilistic model distinguishes the proposed method from a simplex EM algorithm. The spatial hyperparameter vector of $\boldsymbol{\gamma}$ is assumed to be non-informative during model initialization, but theoretical analyses have shown that, the combination of the non-informative hyperparameters and the Gaussian distribution of the signal waveforms results in strong sparsity-promotion [36], [42]. Thus most of the elements in $\boldsymbol{\gamma}$ are forced to converge to 0, and the final reconstruction result contains much limited signal components. That is to say, the proposed method extracts the required information via model shrinkage, while the standard EM algorithm focuses on model refinement with fixed dimension [37].

As only local convergence is guaranteed for the EM iterations, the reconstruction result may not correspond to the global maximum of the likelihood function given in (10). When the likelihood function contains multiple local maxima, the particular maximum that the EM iteration process arrives at depends on how the algorithm is initialized. In this paper, we assume no prior information about the signal, noise or array calibration error, and set the initial values of the hyperparameters as $\boldsymbol{\gamma} = \mathbf{1}_{L \times 1}$, $\mathbf{c} = \mathbf{0}_{P \times 1}$ and $\sigma^2 = 0$. It is very difficult to analyze theoretically that, when the EM algorithm is initialized like this, what the distance is between the reconstruction result and the global maxima of the likelihood function in (10). Actually, it is still an open question to clarify the convergence behavior of the EM algorithm even when the model is calibrated

accurately. But fortunately, the numerical examples in the previous literatures [29], [30] and those to be shown in this paper demonstrate empirically that, the SBL-based DOA estimators work very well when only non-informative prior assumptions are introduced for the hyperparameters. Those empirical results indicate that, the initialization is reasonable and the EM algorithm has good convergence properties when applied in the array signal processing area. If enhanced global convergence performance is required for the reconstruction process, multiple initial hyperparameters can be chosen and the reconstruction result that maximizes the likelihood function should be selected. Alternatively, any reasonable prior information about the signal, noise or array calibration error may be introduced to refine the non-informative initialization and improve the global convergence of the method.

C. Refined DOA Estimation

The hyperparameter set $\boldsymbol{\gamma}$ describes the spatial power distribution of the incident signals on the predefined direction grid, its estimate generally contains K significant peaks when the EM iteration is terminated due to the inner sparsity promotion of the hierarchical probabilistic model [36], [42], while the pseudo-peaks can be eliminated according to the information criterion [30]. Due to the discreteness of $\boldsymbol{\Theta}$, the signal peaks contained in the reconstructed spectrum usually consist of more than one spectral line, and the spatially adjacent lines represent the corresponding signal component in the likelihood function in (10) as a whole. In order to remove the quantization errors in the source locations caused by the model discreteness, we propose in this subsection to replace the adjacent lines with a single line to refine the DOA estimates based on the EM reconstruction result.

When the EM iteration is terminated, denote the estimates of $\mathbf{c}$, σ^2 and $\boldsymbol{\gamma}$ by $\hat{\mathbf{c}}$, $\hat{\sigma}^2$ and $\hat{\boldsymbol{\gamma}}$, respectively, and substitute each signal peak in $\hat{\boldsymbol{\gamma}}$ with a single spectral line, then refined DOA estimates can be obtained by optimizing the locations of those lines. As the SBL reconstruction process well captures local signal properties [39], the refined DOA estimation procedure can be implemented in turn for the K signals. Take the k th signal for example, denote the direction set associated with its spectrum peak by $\boldsymbol{\theta}_k$, the corresponding hyperparameter set by $\hat{\boldsymbol{\gamma}}_k$, $\hat{\boldsymbol{\Gamma}}_k = \text{diag}(\hat{\boldsymbol{\gamma}}_k)$, $\hat{\mathbf{A}}'(\boldsymbol{\theta}_k) = [\hat{\mathbf{a}}'(\boldsymbol{\theta})]_{\boldsymbol{\theta} \in \boldsymbol{\theta}_k}$ with $\hat{\mathbf{a}}'(\boldsymbol{\theta})$ being the array responding vector calibrated with $\hat{\mathbf{c}}$. The variables irrelevant with this signal are identified with subscript $(\bullet)_{-k}$, e.g., $\boldsymbol{\Theta}_{-k} = \boldsymbol{\Theta} \setminus \boldsymbol{\theta}_k$ with “\” meaning removing $\boldsymbol{\theta}_k$ from $\boldsymbol{\Theta}$, $\hat{\boldsymbol{\gamma}}_{-k} = \hat{\boldsymbol{\gamma}} \setminus \hat{\boldsymbol{\gamma}}_k$, $\hat{\boldsymbol{\Gamma}}_{-k} = \text{diag}(\hat{\boldsymbol{\gamma}}_{-k})$, $\hat{\Sigma}_{\mathbf{x}, -k} = \hat{\sigma}^2 \mathbf{I}_M + \hat{\mathbf{A}}'(\boldsymbol{\Theta}_{-k}) \hat{\boldsymbol{\Gamma}}_{-k} (\hat{\mathbf{A}}'(\boldsymbol{\Theta}_{-k}))^H$. Substituting $\hat{\mathbf{A}}'(\boldsymbol{\theta}_k) \hat{\boldsymbol{\Gamma}}_k (\hat{\mathbf{A}}'(\boldsymbol{\theta}_k))^H$ with $\eta \hat{\mathbf{a}}'(\boldsymbol{\theta}) (\hat{\mathbf{a}}'(\boldsymbol{\theta}))^H$ to represent the k th signal component in $\Sigma_{\mathbf{x}}$ yields another formulation of the reconstructed array covariance matrix, i.e., $\Sigma_{\mathbf{x}} = \hat{\Sigma}_{\mathbf{x}, -k} + \eta \hat{\mathbf{a}}'(\boldsymbol{\theta}) (\hat{\mathbf{a}}'(\boldsymbol{\theta}))^H$, then the power and direction of this signal, denoted by η_k and ϑ_k , can be estimated by maximizing (10), i.e.,

$$\begin{aligned} & [\hat{\eta}_k, \hat{\vartheta}_k] \\ &= \arg \max_{\eta, \boldsymbol{\theta}} \left| \pi \left[\hat{\Sigma}_{\mathbf{x}, -k} + \eta \hat{\mathbf{a}}'(\boldsymbol{\theta}) (\hat{\mathbf{a}}'(\boldsymbol{\theta}))^H \right] \right|^{-N} \\ & \quad \times \exp \left\{ -N \text{tr} \left(\left[\hat{\Sigma}_{\mathbf{x}, -k} + \eta \hat{\mathbf{a}}'(\boldsymbol{\theta}) (\hat{\mathbf{a}}'(\boldsymbol{\theta}))^H \right]^{-1} \hat{\mathbf{R}}_{\mathbf{x}} \right) \right\}. \end{aligned} \quad (19)$$

After some straightforward derivation (with details given in Appendix B), the following refined DOA estimator can be obtained,

$$\hat{\theta}_k = \arg \max_{\theta \in \Omega_k} \left| \text{Re} \left\{ (\hat{\mathbf{a}}'(\theta))^H \hat{\Sigma}_{\mathbf{x},-k}^{-1} \left[\hat{\mathbf{a}}'(\theta) (\hat{\mathbf{a}}'(\theta))^H \hat{\Sigma}_{\mathbf{x},-k}^{-1} \hat{\mathbf{R}}_{\mathbf{x}} - \hat{\mathbf{R}}_{\mathbf{x}} \hat{\Sigma}_{\mathbf{x},-k}^{-1} \hat{\mathbf{a}}'(\theta) (\hat{\mathbf{a}}'(\theta))^H \right] \hat{\Sigma}_{\mathbf{x},-k}^{-1} \hat{\mathbf{d}}'(\theta) \right\} \right|^{-1}, \quad (20)$$

where Ω_k represents the space occupied by θ_k . Setting $k = 1, \dots, K$ in (20) obtains the refined DOA estimates of all the signals. Equation (20) has a similar formulation as the refined DOA estimator in accurately calibrated arrays [30], with differences lying in the structures of the variables of $\hat{\Sigma}_{\mathbf{x},-k}$, $\hat{\mathbf{a}}'(\theta)$ and $\hat{\mathbf{d}}'(\theta)$, which depend on the array perturbation error estimate $\hat{\mathbf{c}}$ in (20).

IV. CRAMER-RAO LOWER BOUNDS FOR ARRAY CALIBRATION AND DOA ESTIMATION

The observation model and proposed method for array calibration and DOA estimation in this paper fall into the Bayesian framework, and power-like parameters of γ are introduced to describe the incident signals, thus unconditional CRLB fits the task of evaluating the effectiveness of the proposed method better than the conditional one¹.

The distribution of $\mathbf{x}(t)$ in the case of stochastic signals is

$$\mathbf{x}(t) \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_{\mathbf{x}}), \quad (21)$$

where $\mathbf{R}_{\mathbf{x}} = \lim_{N \rightarrow +\infty} \hat{\mathbf{R}}_{\mathbf{x}} = \mathbf{A}'(\boldsymbol{\vartheta}) \mathbf{R}_{\mathbf{s}} (\mathbf{A}'(\boldsymbol{\vartheta}))^H + \sigma^2 \mathbf{I}_M$ and $\mathbf{R}_{\mathbf{s}} = \lim_{N \rightarrow +\infty} \frac{1}{N} \sum_{n=1}^N \mathbf{s}(t_n) \mathbf{s}^H(t_n)$ [43]. Both real and complex variables are contained in the likelihood function, real ones include $\boldsymbol{\vartheta}$ and σ^2 , while complex ones correspond to the elements of $\mathbf{R}_{\mathbf{s}}$. The array perturbation vector $\mathbf{c}$ is complex in the cases of mutual coupling and gain/phase uncertainty, and is real in the case of sensor location error. We take $\mathbf{c}$ as complex for default during the following analyses for notational convenience, and distinguish the effects of different perturbation types afterwards. The complex variables of $\mathbf{R}_{\mathbf{s}}$ and $\mathbf{c}$ can be decomposed into the sum of real and imaginary parts, i.e., $\mathbf{R}_{\mathbf{s}} = \text{Re}(\mathbf{R}_{\mathbf{s}}) + j\text{Im}(\mathbf{R}_{\mathbf{s}})$ and $\mathbf{c} = \mathbf{c}_R + j\mathbf{c}_I$. As $\mathbf{R}_{\mathbf{s}}$ is a conjugate matrix, the complete parameter set that $\mathbf{x}(t)$ depends on is $\Xi = [\boldsymbol{\vartheta}^T, \mathbf{c}_R^T, \mathbf{c}_I^T, \sigma^2, \{\text{Re}(\mathbf{R}_{\mathbf{s}})_{k_1, k_2}, \text{Im}(\mathbf{R}_{\mathbf{s}})_{k_1, k_2}\}_{1 \leq k_2 < k_1 \leq K}, \{(\mathbf{R}_{\mathbf{s}})_{k, k}\}_{1 \leq k \leq K}]^T$.

The log-likelihood function of $\mathbf{x}(1), \dots, \mathbf{x}(N)$ w.r.t. Ξ is

$$\mathcal{L}(\Xi) = N \ln |\mathbf{R}_{\mathbf{x}}| + N \text{tr} \left(\mathbf{R}_{\mathbf{x}}^{-1} \hat{\mathbf{R}}_{\mathbf{x}} \right). \quad (22)$$

The first-order partial derivations of $\mathcal{L}(\Xi)$ w.r.t. Ξ_i , i.e., the i th element of Ξ , can be computed straightforwardly as

$$\frac{\partial \mathcal{L}(\Xi)}{\partial \Xi_i} = N \text{tr} \left(\mathbf{R}_{\mathbf{x}}^{-1} \frac{\partial \mathbf{R}_{\mathbf{x}}}{\partial \Xi_i} \right) - N \text{tr} \left(\mathbf{R}_{\mathbf{x}}^{-1} \frac{\partial \mathbf{R}_{\mathbf{x}}}{\partial \Xi_i} \mathbf{R}_{\mathbf{x}}^{-1} \hat{\mathbf{R}}_{\mathbf{x}} \right), \quad (23)$$

¹The conditional CRLB was derived in the previous version of this paper, it separates the precision bounds of array perturbation parameters and DOA estimates, but as it serves for the determined (and unknown) signals and is not consistent with the overall Bayesian framework, it is not contained here and left over for another paper.

and the expectation of the second-order derivation of $\mathcal{L}(\Xi)$ w.r.t. Ξ_{i_1} and Ξ_{i_2} is

$$\mathbb{E} \left\{ \frac{\partial^2 \mathcal{L}(\Xi)}{\partial \Xi_{i_1} \partial \Xi_{i_2}} \right\} = N \text{tr} \left(\mathbf{R}_{\mathbf{x}}^{-1} \frac{\partial \mathbf{R}_{\mathbf{x}}}{\partial \Xi_{i_1}} \mathbf{R}_{\mathbf{x}}^{-1} \frac{\partial \mathbf{R}_{\mathbf{x}}}{\partial \Xi_{i_2}} \right), \quad (24)$$

The expressions of $\partial \mathbf{R}_{\mathbf{x}} / \partial \Xi_i$ for different Ξ_i can be derived directly based on $\mathbf{R}_{\mathbf{x}} = \mathbf{A}'(\boldsymbol{\vartheta}) \mathbf{R}_{\mathbf{s}} (\mathbf{A}'(\boldsymbol{\vartheta}))^H + \sigma^2 \mathbf{I}_M$. Thus the CRLB of Ξ satisfies

$$[\text{CRLB}_{\Xi}^{-1}]_{i_1, i_2} = N \text{tr} \left(\mathbf{R}_{\mathbf{x}}^{-1} \frac{\partial \mathbf{R}_{\mathbf{x}}}{\partial \Xi_{i_1}} \mathbf{R}_{\mathbf{x}}^{-1} \frac{\partial \mathbf{R}_{\mathbf{x}}}{\partial \Xi_{i_2}} \right). \quad (25)$$

The parameters of $\mathbf{R}_{\mathbf{s}}$ and σ^2 are nuisance ones during DOA estimation and array calibration, they can be substituted with their maximum likelihood (ML) estimates in (22) to simplify the CRLB analysis of $\boldsymbol{\vartheta}$ and $\mathbf{c}$. The subjective function of $\Xi' = [\boldsymbol{\vartheta}^T, \mathbf{c}_R^T, \mathbf{c}_I^T]^T$ after the substitution is

$$\mathcal{L}(\Xi') = \ln |\mathbf{A}'(\boldsymbol{\vartheta}) \mathbf{Z} (\mathbf{A}'(\boldsymbol{\vartheta}))^H + \kappa \mathbf{I}_M|, \quad (26)$$

where

$$\mathbf{Z} = (\mathbf{A}'(\boldsymbol{\vartheta}))^\dagger \hat{\mathbf{R}}_{\mathbf{x}} [(\mathbf{A}'(\boldsymbol{\vartheta}))^\dagger]^H - \kappa [(\mathbf{A}'(\boldsymbol{\vartheta}))^H \mathbf{A}'(\boldsymbol{\vartheta})]^{-1}, \quad (27)$$

$$\kappa = \frac{1}{M - K} \text{tr} \left\{ \mathbf{P}_{\mathbf{A}'(\boldsymbol{\vartheta})}^\perp \hat{\mathbf{R}}_{\mathbf{x}} \right\}, \quad (28)$$

and $\mathbf{P}_{\mathbf{A}'(\boldsymbol{\vartheta})}^\perp = \mathbf{I}_M - \mathbf{A}'(\boldsymbol{\vartheta}) (\mathbf{A}'(\boldsymbol{\vartheta}))^\dagger$ and $(\mathbf{A}'(\boldsymbol{\vartheta}))^\dagger = [(\mathbf{A}'(\boldsymbol{\vartheta}))^H \mathbf{A}'(\boldsymbol{\vartheta})]^{-1} (\mathbf{A}'(\boldsymbol{\vartheta}))^H$ [43].

Denote

$$\hat{\Xi}' = \arg \max_{\Xi'} \mathcal{L}(\Xi'), \quad (29)$$

then the CRLB of Ξ' can be derived according to $\mathbb{E}\{(\hat{\Xi}' - \Xi')(\hat{\Xi}' - \Xi')^T\}$ [43], [44]. Expanding $\partial \mathcal{L}(\hat{\Xi}') / \partial \Xi' = \mathbf{0}$ in the neighborhood of Ξ' and neglecting the second (and higher) order items yields

$$\frac{\partial \mathcal{L}(\Xi')}{\partial \Xi'} + \frac{\partial^2 \mathcal{L}(\Xi')}{(\partial \Xi')^2} (\hat{\Xi}' - \Xi') = 0, \quad (30)$$

thus

$$\begin{aligned} & \mathbb{E} \left\{ (\hat{\Xi}' - \Xi') (\hat{\Xi}' - \Xi')^T \right\} \\ &= \mathbb{E} \left\{ \left[\frac{\partial^2 \mathcal{L}(\Xi')}{(\partial \Xi')^2} \right]^{-1} \frac{\partial \mathcal{L}(\Xi')}{\partial \Xi'} \left(\frac{\partial \mathcal{L}(\Xi')}{\partial \Xi'} \right)^T \left[\frac{\partial^2 \mathcal{L}(\Xi')}{(\partial \Xi')^2} \right]^{-1} \right\}. \end{aligned} \quad (31)$$

Following the derivation guidelines in [43], the CRLB of Ξ' can be obtained based on (26) as

$$[\text{CRLB}_{\Xi'}^{-1}]_{i_1, i_2} = \frac{2N}{\sigma^2} \text{Re} \left\{ \text{tr} \left[\mathbf{U} \mathbf{W}^{(i_1, i_2)} \right] \right\}, \quad (32)$$

where $\mathbf{U} = \mathbf{R}_{\mathbf{s}} (\mathbf{A}'(\boldsymbol{\vartheta}))^H \mathbf{R}_{\mathbf{x}}^{-1} \mathbf{A}'(\boldsymbol{\vartheta}) \mathbf{R}_{\mathbf{s}}$ and $\mathbf{W}^{(i_1, i_2)} = (\frac{\partial \mathbf{A}'(\boldsymbol{\vartheta})}{\partial \Xi_{i_2}})^H \mathbf{P}_{\mathbf{A}'(\boldsymbol{\vartheta})}^\perp \frac{\partial \mathbf{A}'(\boldsymbol{\vartheta})}{\partial \Xi_{i_1}}$. In accurately calibrated arrays, $\partial \mathbf{A}(\boldsymbol{\vartheta}) / \partial \Xi'_i$ contains only one nonzero column, and $\text{CRLB}_{\Xi'}^{-1}$ can be written in a more compact form [43], but the inclusion of the array perturbation parameters invalidates the simplification process.

The CRLB of $\boldsymbol{\vartheta}$, $\mathbf{c}_R$ and $\mathbf{c}_I$ can be obtained by partitioning the expressions in (25) or (32), which are denoted by $\text{CRLB}_{\boldsymbol{\vartheta}}$,

CRLB_{c_R} and CRLB_{c_I} for notational convenience. In the aforementioned results, the perturbation errors are decomposed into somewhat meaningless real and imaginary parts in the case of mutual coupling and gain/phase uncertainty to facilitate derivation, thus the precisions of those perturbation parameter estimates are not directly available. The results in [45] can be exploited to transform the CRLB of $[\mathbf{c}_R^T, \mathbf{c}_I^T]^T$ to that of $[\mathbf{c}^T, \mathbf{c}^H]^T$ to give a better description of the upper precision bound of $\mathbf{c}$. After the transformation, the lower bounds of the estimation precision of each element of $\mathbf{c}$ and the whole vector are given by

$$\mathbb{E}\{(\hat{c}_i - c_i)(\hat{c}_i - c_i)^*\} \geq (\text{CRLB}_{\mathbf{c}_R})_{i,i} + (\text{CRLB}_{\mathbf{c}_I})_{i,i}, \quad (33)$$

$$\mathbb{E}\{(\hat{\mathbf{c}} - \mathbf{c})^H(\hat{\mathbf{c}} - \mathbf{c})\} \geq \text{tr}(\text{CRLB}_{\mathbf{c}_R} + \text{CRLB}_{\mathbf{c}_I}). \quad (34)$$

In the presence of sensor location errors, $\mathbf{c}_I = \emptyset$, (33) and (34) can be simplified as $\mathbb{E}\{(\hat{c}_i - c_i)(\hat{c}_i - c_i)^*\} \geq (\text{CRLB}_{\mathbf{c}_R})_{i,i}$ and $\mathbb{E}\{(\hat{\mathbf{c}} - \mathbf{c})^H(\hat{\mathbf{c}} - \mathbf{c})\} \geq \text{tr}(\text{CRLB}_{\mathbf{c}_R})$, respectively.

V. DOA ESTIMATION WHEN MULTIPLE TYPES OF PERTURBATION ERRORS COEXIST

In this section, we take the scenario when both mutual coupling and gain/phase uncertainty exist for example, so as to show how the method proposed in Section III can be extended to solve the DOA estimation problem when multiple types of perturbation errors coexist. The array output at time t in the particular scenario is

$$\begin{aligned} \mathbf{x}(t) &= \mathbf{C}^{(I)} \mathbf{C}^{(II)} \mathbf{A}(\boldsymbol{\theta}) \mathbf{s}(t) + \mathbf{v}(t) \\ &= \mathbf{A}'(\boldsymbol{\theta}) \mathbf{s}(t) + \mathbf{v}(t), \end{aligned} \quad (35)$$

where $\mathbf{C}^{(I)} = \text{toeplitz}([1, b_1, \dots, b_P, \mathbf{0}_{(M-P-1) \times 1}^T])^T$ and $\mathbf{C}^{(II)} = \text{diag}([1, \alpha_2 e^{j\phi_2}, \dots, \alpha_M e^{j\phi_M}]^T)$ are the mutual coupling and gain-phase matrices, respectively. Equation (35) can be transformed as follows to make clearer the relationships between the array output and the perturbation parameters,

$$\begin{aligned} \mathbf{x}(t) &= \mathbf{C}^{(II)} \mathbf{A}(\boldsymbol{\theta}) \mathbf{s}(t) + \mathbf{Q}_t^{(I)} \mathbf{c}^{(I)} + \mathbf{v}(t) \\ &= \mathbf{C}^{(I)} \mathbf{A}(\boldsymbol{\theta}) \mathbf{s}(t) + \mathbf{Q}_t^{(II)}(t) \mathbf{c}^{(II)} + \mathbf{v}(t), \end{aligned} \quad (36)$$

where $[\mathbf{Q}_t^{(I)}]_{:,p} = \frac{\partial}{\partial c_p^{(I)}}[(\mathbf{C}^{(I)} - \mathbf{I}_M) \mathbf{C}^{(II)} \mathbf{A}(\boldsymbol{\theta}) \mathbf{s}(t)]$ and $[\mathbf{Q}_t^{(II)}]_{:,p} = \frac{\partial}{\partial c_p^{(II)}}[\mathbf{C}^{(I)} (\mathbf{C}^{(II)} - \mathbf{I}_M) \mathbf{A}(\boldsymbol{\theta}) \mathbf{s}(t)]$.

Extending (35) and (36) from $\boldsymbol{\theta}$ to Θ via zero-padding yields the overcomplete formulation of the array output, and the SBL technique [36]–[39] can be introduced to reconstruct the spatial source distribution and estimate the perturbation parameters. The coexistence of multiple types of perturbation errors does not change the basic guideline of the Bayesian reconstruction procedure, but the iteration strategy of the EM algorithm should be modified to take all the perturbations into account. The modified M-step contains the following updates,

$$\begin{aligned} (\mathbf{c}^{(I)})^{(q)} &= \left\langle \sum_{t=1}^N (\bar{\mathbf{Q}}_t^{(I)})^H \bar{\mathbf{Q}}_t^{(I)} \right\rangle^{-1} \\ &\times \left\langle \sum_{t=1}^N (\bar{\mathbf{Q}}_t^{(I)})^H (\mathbf{x}(t) - \mathbf{C}^{(II)} \mathbf{A}(\Theta) \bar{\mathbf{s}}(t)) \right\rangle, \end{aligned} \quad (37)$$

$$\begin{aligned} (\mathbf{c}^{(II)})^{(q)} &= \left\langle \sum_{t=1}^N (\bar{\mathbf{Q}}_t^{(II)})^H \bar{\mathbf{Q}}_t^{(II)} \right\rangle^{-1} \\ &\times \left\langle \sum_{t=1}^N (\bar{\mathbf{Q}}_t^{(II)})^H (\mathbf{x}(t) - \mathbf{C}^{(I)} \mathbf{A}(\Theta) \bar{\mathbf{s}}(t)) \right\rangle, \end{aligned} \quad (38)$$

$$(\sigma^2)^{(q)} = \frac{1}{MN} \left\langle \sum_{t=1}^N \|\mathbf{x}(t) - (\mathbf{A}'(\Theta))^{(q)} \bar{\mathbf{s}}(t)\|_2^2 \right\rangle, \quad (39)$$

$$\gamma_l^{(q)} = \frac{1}{N} \left\langle \sum_{t=1}^N |\bar{s}_l(t)|^2 \right\rangle, \quad (40)$$

the variables in (37)–(40) are defined similarly as their counterparts in Section III, $\bar{\mathbf{s}}(t) | \mathbf{x}(t) \sim \mathcal{N}(\boldsymbol{\mu}_t, \boldsymbol{\Sigma}_{\bar{\mathbf{s}}})$ with $\boldsymbol{\mu}_t$ and $\boldsymbol{\Sigma}_{\bar{\mathbf{s}}}$ given by (45) and (44), respectively, and $\mathbf{A}'(\Theta)$ is the zero-padded extension of $\mathbf{A}'(\boldsymbol{\theta})$ in (35). Similar derivations as those in (46)–(49) can be carried out to transform (37)–(40) to $\boldsymbol{\mu}_t$ - and $\boldsymbol{\Sigma}_{\bar{\mathbf{s}}}$ -dependent formulations, and details are left out here for conciseness. When the EM algorithm is terminated, the method presented in Section III-C can be used to obtain refined DOA estimates.

VI. NUMERICAL EXAMPLES

In this part, we carry out simulations to demonstrate the performance of the proposed DOA estimation method, named Sparse Bayesian Array Calibration and SBAC for short, single-perturbation-type scenarios are considered in Subsection A–D, while a multiple-perturbation-type scenario is considered in Subsection E. Two equal-power independent signals are supposed to impinge onto an 8-element linear array, and the nominal geometry of the array is uniform with inter-element spacing equaling half-wavelength of the incident signals. The $[-90^\circ, 90^\circ]$ space is sampled with 1° interval to form the direction set of $\Theta = [-90^\circ, -89^\circ, \dots, 90^\circ]$ in SBAC, and the EM algorithm is terminated when the update ratio of $\boldsymbol{\gamma}$ is smaller than 1×10^{-4} in two successive iterations, i.e., $\|\boldsymbol{\gamma}^{(q+1)} - \boldsymbol{\gamma}^{(q)}\|_2 / \|\boldsymbol{\gamma}^{(q)}\|_2 < 10^{-4}$. The signal number is assumed to be known beforehand, and the required DOA estimation precision is $(10^{-(\text{SNR}/20 - 1)})^\circ$ for all the methods, which depends on the SNR of the incident signals and is about 1/10 of the CRLB, which are computed according to (25) in the simulations. The average root-mean-square error (RMSE) of the DOA estimates of the two signals is used for statistical direction estimation precision evaluation, which is defined as

$$\text{RMSE}_{\boldsymbol{\theta}} = \sqrt{\sum_{i=1}^I \left\| \hat{\boldsymbol{\theta}}^{(i)} - \boldsymbol{\theta}^{(i)} \right\|_2^2 / (K \times I)}, \quad (41)$$

where I denotes the number of simulations in a given scenario and we set $I = 300$ in subsection B–D, $\boldsymbol{\theta}^{(i)}$ and $\hat{\boldsymbol{\theta}}^{(i)}$ are the true and estimated directions in the i th simulation, and $\boldsymbol{\theta}^{(i)}$ changes with a small magnitude in different simulations. Similarly, the RMSE of the array calibration precision defined below is evaluated,

$$\text{RMSE}_{\mathbf{c}} = \varsigma \times \sqrt{\sum_{i=1}^I \left\| \hat{\mathbf{c}}^{(i)} - \mathbf{c} \right\|_2^2 / I}, \quad (42)$$

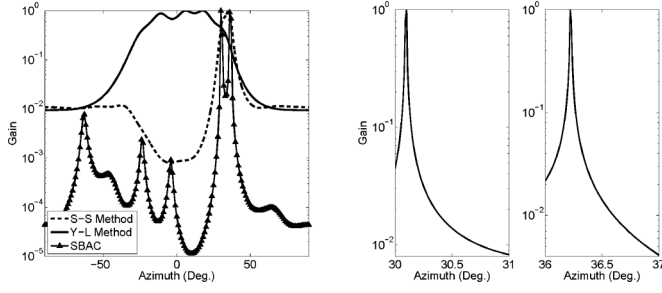


Fig. 1. Spectra in a mutually coupled array, Left: the SBAC spectrum is obtained on Θ ; Right: refined SBAC spectra.

where c keeps constant in each scenario, ς is a tuning factor introduced to enhance the sense of the RMSE, $\varsigma = \|c\|_2$ for mutual coupling, $\varsigma = 1$ for gain/phase uncertainty and $\varsigma = \lambda^{-1}$ for sensor location error.

A. Spectra of SBAC

Suppose that two 10 dB signals impinge onto the array from directions of 30.3° and 36.8° , the mutual coupling effect is significant and the coupling coefficient vector is $c = [0.6 + j0.4, 0.2 + j0.1]^T$. Ten snapshots are collected by the array, and the methods of SBAC, S-S Method [2] and Y-L Method [3] are used to estimate the source directions. The spectra of those methods given in the first figure in Fig. 1 indicate that, only SBAC is able to separate the two signals, and the perturbation parameter estimate is $\hat{c} = [0.576 + j0.374, 0.225 + j0.109]^T$, whose relative error is only 5.9%. One can also conclude from the SBAC spectrum on the direction grid Θ that, the spectrum peaks are very sharp and each peak generally consists of two adjacent spectral lines with significantly nonzero magnitudes. Therefore, during the refined DOA estimation procedure, only those two lines are retained to form the corresponding signal peak, and the scanning space in (20) is the 1° scope between the two lines. The refined SBAC spectra for the two sources are listed in the second figure in Fig. 1, and the DOA estimates derived from them are 30.10° and 36.22° , respectively, with estimation errors smaller than 0.6° . This group of simulation results intuitively demonstrates the predominance of SBAC over the existing counterparts in array calibration and DOA estimation performance.

B. Statistical Performance in the Presence of Mutual Coupling

Suppose that the two sources are separated by 10° , and the array manifold is perturbed by the mutual coupling effect with the same coefficients as those in Subsection A. The directions of the two sources are $30^\circ + \zeta$ and $40^\circ + \zeta$, respectively, with ζ set randomly and uniformly within $[-0.5^\circ, 0.5^\circ]$ to remove the possible prior directional information contained in the predefined grid set Θ . The snapshot number is fixed at 30 and the SNR of the two signals varies from -3 dB to 20 dB, the DOA estimation and array calibration RMSE of SBAC, S-S Method and Y-L Method are shown in Fig. 2(a) and (b). The precisions of SBAC in DOA estimation and array calibration are very close to the CRLB when the SNR is only 0 dB, and they improve with a similar speed as that of the CRLB when the SNR increases on. Y-L Method uses the output only on the central 4-element sub-array when estimating the mutual coupling coefficients, thus it fails to achieve satisfying array calibration and DOA estimation

precisions, even when the SNR is as high as 20 dB. S-S Method succeeds to separate the two sources when the SNR exceeds 5 dB, and it arrives at high-precision array calibration results simultaneously, but its DOA estimation and array calibration precisions are lower than those of SBAC all along.

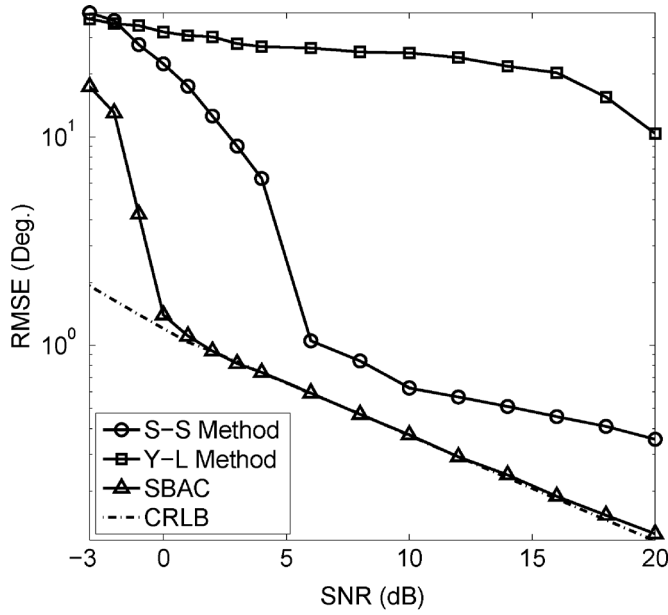
Then fix the SNR of the two signals at 10 dB and set the source directions according to the same strategy as that in the simulations associated with Fig. 2, the DOA estimation and array calibration RMSE of SBAC, S-S Method and Y-L Method when the snapshot number increases from 2 to 1024 are shown in Fig. 3(a) and (b). SBAC surpasses the other two methods significantly in both limited snapshot adaptation and parameter estimation precision, it needs only 8 snapshots to obtain satisfying DOA and mutual coupling parameter estimates with precisions very close to the CRLB. S-S Method needs more than 16 snapshots to separate the two sources, and its parameter estimation precisions depart from the CRLB with obvious margins. Y-L Method again behaves the worst, it fails to obtain satisfying DOA estimation and array calibration results even when as many as 1024 snapshots are collected.

C. Statistical Performance in the Presence of Gain/Phase Uncertainty

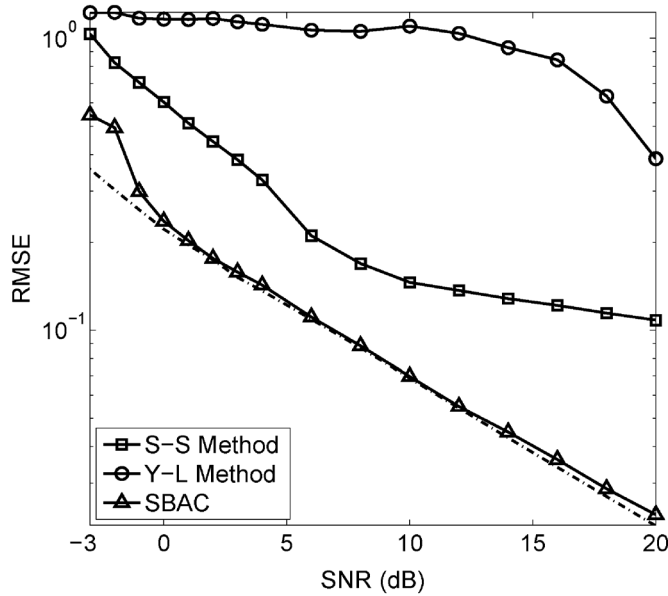
In this subsection, we suppose that the array manifold is perturbed by gain and phase uncertainties, the gain errors of the sensors are 0, 0.15, 0.3, 0.2, 0.25, -0.2 , -0.2 and -0.3 in turn, and the phase errors are 0° , -20° , -30° , -40° , 35° , 25° , 20° and 40° . Two signals departed by 10° , with their directions set according to the same strategy as that in the simulations in Subsection VI-B, impinge onto the array simultaneously, and 30 snapshots are collected. When the SNR increases from -5 dB to 20 dB, the DOA estimation and array calibration RMSE of SBAC, Eigenstructure Method [5] and L-E Method [6] are given in Fig. 4(a) and (b). As the gain and phase uncertainties are very significant, Eigenstructure Method and L-E Method do not perform very well in array calibration all along, and the DOA estimation precisions of them are still about 3° when the signal SNR is as high as 20 dB. The DOA estimation RMSE of SBAC is smaller than 3° when the SNR exceeds 0 dB, and it decreases rapidly as the SNR increases on. The gain and phase estimation RMSE of SBAC is also much smaller than those of Eigenstructure Method and L-E Method, and it improves with a similar trend as that of the SBAC DOA estimates. However, as the effect of gain and phase uncertainties are much more difficult to separate from the array manifold than the mutual coupling effect, the DOA estimation and array calibration precisions of SBAC fail to reach the CRLB at high SNR.

D. Statistical Performance in the Presence of Sensor Location Error

Suppose that the sensor locations are not accurately calibrated in this subsection, they depart from the nominal positions by 0, 0.08, 0.12, 0.16, 0.04, -0.12 , -0.08 and -0.16 times the half-wavelength. Two signals separated by 10° impinge onto the array simultaneously, the signal directions are set similarly as that in the simulations according to Fig. 2, and 30 snapshots are collected by the array. When the signal SNR increases from -5 dB to 20 dB, the DOA estimation and array calibration RMSE of SBAC, the maximum likelihood method (denoted by ML Method) [8] and IMTAM [9] are given in Fig. 5(a) and (b).



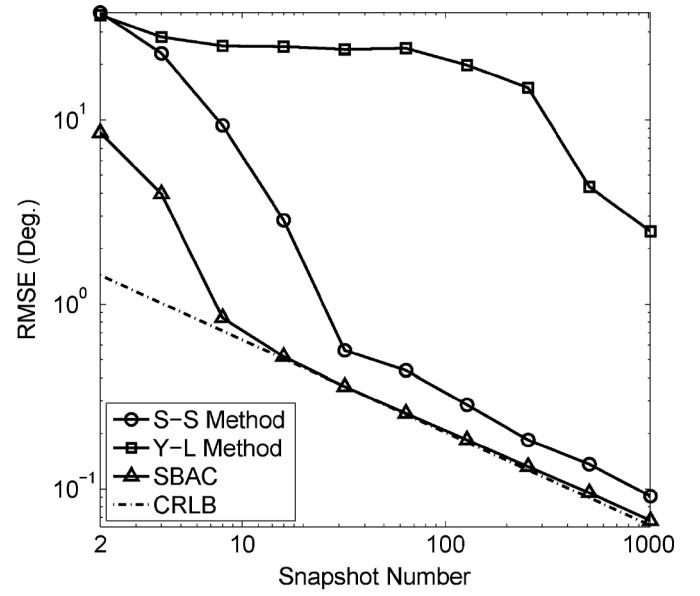
(a)



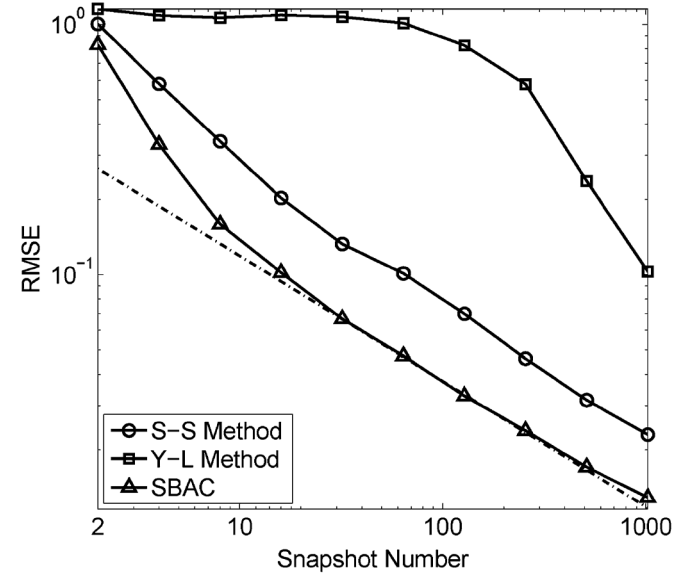
(b)

Fig. 2. DOA estimation and array calibration precision in the presence of mutual coupling at different SNR, (a) DOA estimation; (b) array calibration.

SBAC again shows enhanced low-SNR adaptation and higher parameter estimation precision than the existing methods. Its DOA estimation RMSE is smaller than 2° and very close to the CRLB when the signal SNR exceeds 5 dB. Its array calibration precision also improves at a similar speed as the CRLB. ML Method and IMTAM succeed to separate the two signals when the SNR is higher than 8 dB, but their DOA estimation precisions, together with the array calibration precision of ML Method (no array calibration result is available from IMTAM), improve only slightly for increasing SNR, and their DOA estimation RMSE are still larger than 1° when the SNR is as high as 20 dB.



(a)

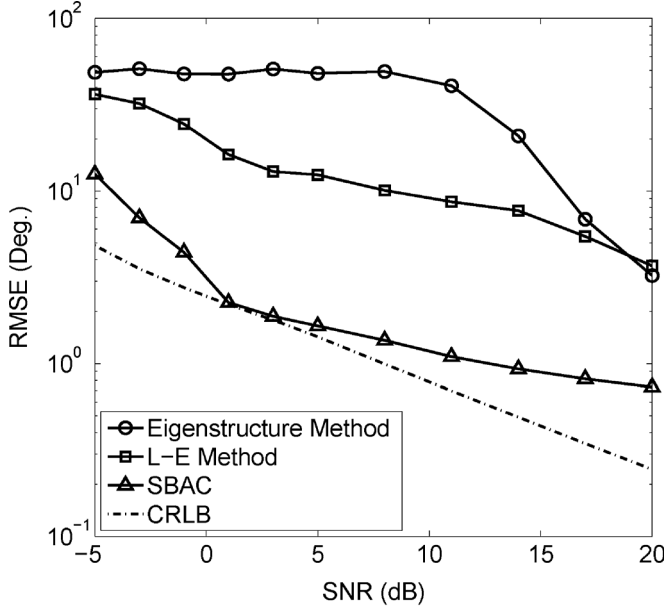


(b)

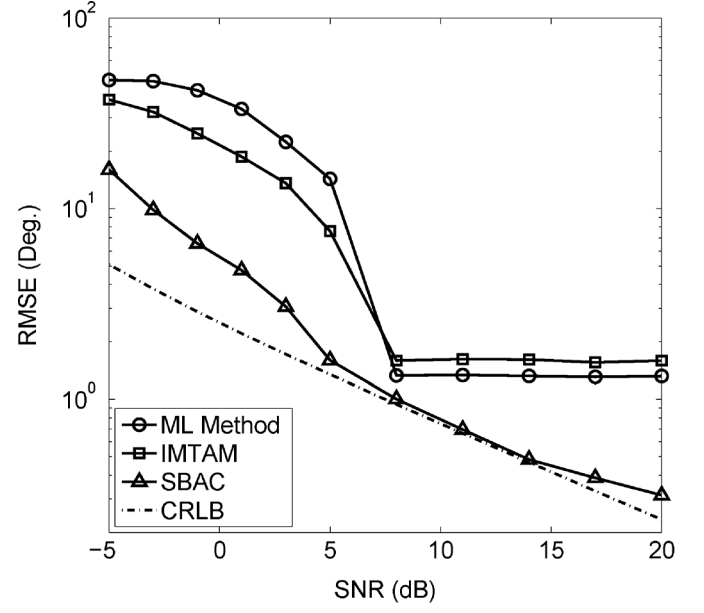
Fig. 3. DOA estimation and array calibration precision in the presence of mutual coupling with different numbers of snapshots, (a) DOA estimation; (b) array calibration.

E. Joint Calibration of Mutual Coupling and Gain/Phase Uncertainty

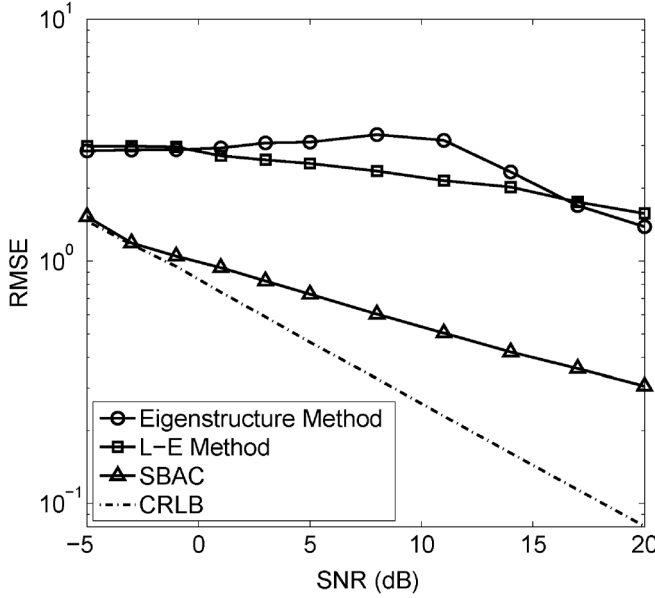
In the following simulation, we suppose that the array manifold is perturbed by both mutual coupling and gain/phase uncertainty, with the perturbation parameters set the same as those in Subsection VI-B and C. Two 20 dB signals impinge onto the array from directions of -10° and 20° simultaneously, and 100 snapshots are collected. The joint calibration method presented in Section V is used to calibrate the array manifold and estimate the source directions. The perturbation parameter estimation results of both types of array imperfections are listed in Tables I and II, respectively, the SBAC spectrum after array calibration is shown in Fig. 6, and the refined DOA estimates



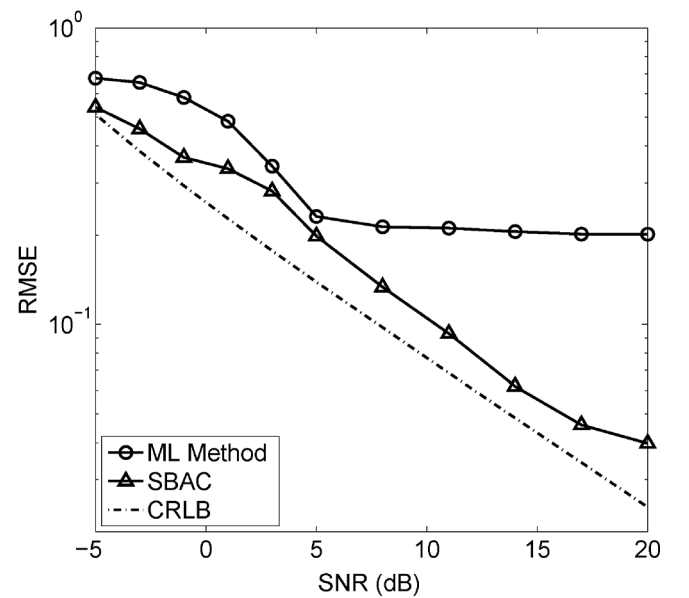
(a)



(a)



(b)



(b)

Fig. 4. DOA estimation and array calibration precision in the presence of gain/phase uncertainty at different SNR, (a) DOA estimation; (b) array calibration.

Fig. 5. DOA estimation and array calibration precision in the presence of sensor location error at different SNR, (a) DOA estimation; (b) array calibration.

of the two sources obtained based on the reconstruction result are -9.02° and 21.03° . It can be concluded from the simulation results that the relative estimation error of the mutual coupling coefficients is as small as 1%, the gain/phase perturbation is also calibrated with high precision, and the DOA estimation errors of both sources are only about 1° . This simulation indicates that SBAC also performs well when multiple types of array perturbation errors coexist.

VII. CONCLUSION

The DOA estimation problem in the presence of mutual coupling, gain/phase uncertainty and sensor location error is studied

in this paper. A comprehensive array output model that is applicable to each type of the typical array perturbations is formulated first, and a sparse Bayesian method named SBAC is proposed for array calibration and DOA estimation based on the comprehensive model. The CRLB of the array calibration and DOA estimation precisions are also derived. The perturbed model, the SBAC method and the CRLB can all be reified to apply in the considered typical perturbation scenarios with a single array imperfection type, and those results can also be extended straightforwardly to other perturbed scenarios, such as mutual coupling compensation in uniform circular arrays. The simulation results show that, SBAC has much enhanced adaptation than the existing methods to demanding scenarios with low

TABLE I
MUTUAL COUPLING CALIBRATION RESULT

	True	Estimate
$c_1^{(I)}$	0.6+j0.4	0.602+j0.402
$c_2^{(I)}$	0.2+j0.1	0.193+j0.101

TABLE II
GAIN/PHASE UNCERTAINTY CALIBRATION RESULT

	True	Estimate
$c_1^{(II)}$	1.081-j0.393	1.014-j0.434
$c_2^{(II)}$	1.126-j0.650	1.000-j0.732
$c_3^{(II)}$	0.919-j0.771	0.748-j0.884
$c_4^{(II)}$	1.024+j0.717	1.128+j0.464
$c_5^{(II)}$	0.725+j0.338	0.760+j0.130
$c_6^{(II)}$	0.752+j0.274	0.761+j0.021
$c_7^{(II)}$	0.536+j0.450	0.651+j0.206

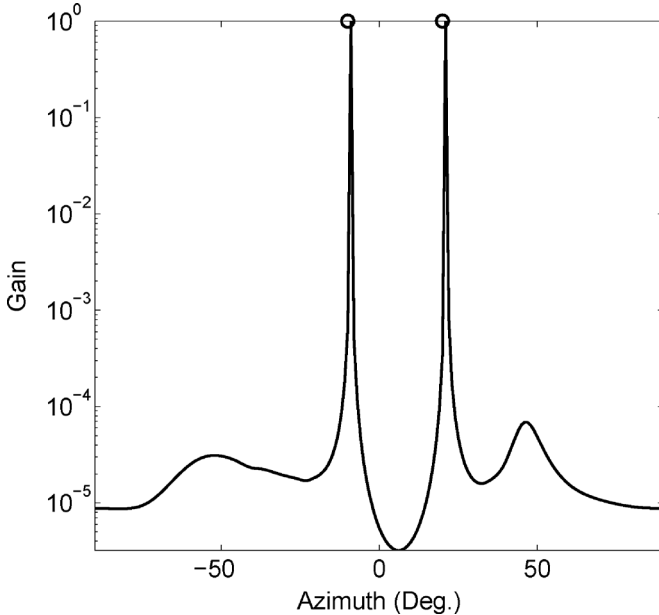


Fig. 6. SBAC spectrum after joint mutual coupling and gain/phase calibration.

SNR, limited snapshots and spatially adjacent signals, it also obtains higher DOA estimation and array calibration precisions. SBAC has also been demonstrated to be extensible to solve the DOA estimation problem when multiple types of array imperfections coexist.

APPENDIX A IMPLEMENTATION OF THE EM ITERATION

The posterior density of $\bar{\mathbf{s}}(t)$ w.r.t. $\{\mathbf{x}(t)\}_{t=1}^N$ can be obtained according to (9) and (8) to be

$$\begin{aligned}
 & p(\{\bar{\mathbf{s}}(t)\}_{t=1}^N | \{\mathbf{x}(t)\}_{t=1}^N; \boldsymbol{\gamma}, \mathbf{c}, \sigma^2) \\
 & \propto p(\{\mathbf{x}(t)\}_{t=1}^N | \{\bar{\mathbf{s}}(t)\}_{t=1}^N; \mathbf{c}, \sigma^2) p(\{\bar{\mathbf{s}}(t)\}_{t=1}^N; \boldsymbol{\gamma}) \\
 & = \prod_{t=1}^N p(\mathbf{x}(t) | \bar{\mathbf{s}}(t); \mathbf{c}, \sigma^2) p(\bar{\mathbf{s}}(t); \boldsymbol{\gamma}) \\
 & = \prod_{t=1}^N |\pi \boldsymbol{\Sigma}_{\bar{\mathbf{s}}}^{-1}| \exp \left\{ -(\bar{\mathbf{s}}(t) - \boldsymbol{\mu}_t)^H \boldsymbol{\Sigma}_{\bar{\mathbf{s}}}^{-1} (\bar{\mathbf{s}}(t) - \boldsymbol{\mu}_t) \right\}, \quad (43)
 \end{aligned}$$

where

$$\begin{aligned}
 \boldsymbol{\Sigma}_{\bar{\mathbf{s}}} &= [\boldsymbol{\Gamma}^{-1} + \sigma^{-2} (\mathbf{A}'(\boldsymbol{\Theta}))^H \mathbf{A}'(\boldsymbol{\Theta})]^{-1} \\
 &= \boldsymbol{\Gamma} - \boldsymbol{\Gamma} (\mathbf{A}'(\boldsymbol{\Theta}))^H \boldsymbol{\Sigma}_{\mathbf{x}}^{-1} \mathbf{A}'(\boldsymbol{\Theta}) \boldsymbol{\Gamma}, \quad (44)
 \end{aligned}$$

$$\boldsymbol{\mu}_t = \sigma^{-2} \boldsymbol{\Sigma}_{\bar{\mathbf{s}}} (\mathbf{A}'(\boldsymbol{\Theta}))^H \mathbf{x}(t) = \boldsymbol{\Gamma} (\mathbf{A}'(\boldsymbol{\Theta}))^H \boldsymbol{\Sigma}_{\mathbf{x}}^{-1} \mathbf{x}(t). \quad (45)$$

Equation (43) indicates that $\bar{\mathbf{s}}(t) | \mathbf{x}(t) \sim \mathcal{N}(\boldsymbol{\mu}_t, \boldsymbol{\Sigma}_{\bar{\mathbf{s}}})$, thus $\mathbf{x}(t) - \mathbf{A}'(\boldsymbol{\Theta}) \bar{\mathbf{s}}(t) \sim \mathcal{N}(\mathbf{x}(t) - \mathbf{A}'(\boldsymbol{\Theta}) \boldsymbol{\mu}_t, \mathbf{A}'(\boldsymbol{\Theta}) \boldsymbol{\Sigma}_{\bar{\mathbf{s}}} (\mathbf{A}'(\boldsymbol{\Theta}))^H)$. Denote $\mathbf{M} = [\boldsymbol{\mu}_1, \dots, \boldsymbol{\mu}_N]$, then the following conclusion can be derived from $[\bar{\mathbf{Q}}_t]_{:,p} = \mathbf{G}_p \boldsymbol{\Phi}(\boldsymbol{\Theta}) \bar{\mathbf{s}}(t)$,

$$\begin{aligned}
 & \left\langle \sum_{t=1}^N \bar{\mathbf{Q}}_t^H \bar{\mathbf{Q}}_t \right\rangle_{p_1, p_2} \\
 &= \left\langle \sum_{t=1}^N \bar{\mathbf{s}}^H(t) \boldsymbol{\Phi}^H(\boldsymbol{\Theta}) \mathbf{G}_{p_1}^H \mathbf{G}_{p_2} \boldsymbol{\Phi}(\boldsymbol{\Theta}) \bar{\mathbf{s}}(t) \right\rangle \\
 &= \text{tr} [\mathbf{G}_{p_1}^H \mathbf{G}_{p_2} \boldsymbol{\Phi}(\boldsymbol{\Theta}) \langle \bar{\mathbf{S}} \bar{\mathbf{S}}^H \rangle \boldsymbol{\Phi}^H(\boldsymbol{\Theta})] \\
 &= \text{tr} [\mathbf{G}_{p_1}^H \mathbf{G}_{p_2} \boldsymbol{\Phi}(\boldsymbol{\Theta}) (\mathbf{M} \mathbf{M}^H + N \boldsymbol{\Sigma}_{\bar{\mathbf{s}}}) \boldsymbol{\Phi}^H(\boldsymbol{\Theta})]. \quad (46)
 \end{aligned}$$

where the subscript $\langle \bullet \rangle_{p_1, p_2}$ indicates the element on the p_1 th row and p_2 th column of the matrix in the angle brackets. Similarly, one can obtain that

$$\left\langle \sum_{t=1}^N \bar{\mathbf{Q}}_t^H \mathbf{x}(t) \right\rangle_p = \text{tr} [\mathbf{G}_p^H \mathbf{X} \mathbf{M}^H \boldsymbol{\Phi}^H(\boldsymbol{\Theta})], \quad (47)$$

$$\begin{aligned}
 & \left\langle \sum_{t=1}^N \bar{\mathbf{Q}}_t^H \mathbf{A}(\boldsymbol{\Theta}) \bar{\mathbf{s}}(t) \right\rangle_p \\
 &= \text{tr} [\mathbf{G}_p^H \mathbf{A}(\boldsymbol{\Theta}) (\mathbf{M} \mathbf{M}^H + N \boldsymbol{\Sigma}_{\bar{\mathbf{s}}}) \boldsymbol{\Phi}^H(\boldsymbol{\Theta})]. \quad (48)
 \end{aligned}$$

where the subscript $\langle \bullet \rangle_p$ indicates the p th element of the vector in the angle brackets. Moreover,

$$\begin{aligned}
 & \left\langle \sum_{t=1}^N \left\| \mathbf{x}(t) - (\mathbf{A}'(\boldsymbol{\Theta}))^{(q)} \bar{\mathbf{s}}(t) \right\|_2^2 \right\rangle \\
 &= \|\mathbf{X} - \mathbf{A}'(\boldsymbol{\Theta}) \mathbf{M}\|_F^2 + N \text{tr} (\mathbf{A}'(\boldsymbol{\Theta}) \boldsymbol{\Sigma}_{\bar{\mathbf{s}}} (\mathbf{A}'(\boldsymbol{\Theta}))^H), \quad (49)
 \end{aligned}$$

where $\bar{\mathbf{S}} = [\bar{\mathbf{s}}(1), \dots, \bar{\mathbf{s}}(N)]$ and $\mathbf{X} = [\mathbf{x}(1), \dots, \mathbf{x}(N)]$.

Substituting the conditional expectations given in (46)–(49) into (16)–(18) yielding more convenient steps for updating $\mathbf{c}$, σ^2 and $\boldsymbol{\gamma}$ and implementing the EM iterations.

APPENDIX B

DERIVATION OF THE REFINED DOA ESTIMATOR

Equation (19) can be rewritten as follows,

$$\begin{aligned} & [\hat{\eta}_k, \hat{\vartheta}_k] \\ &= \arg \min_{\eta, \theta} \left\{ \ln \left| \hat{\Sigma}_{\mathbf{x}, -k} + \eta \hat{\mathbf{a}}'(\theta) (\hat{\mathbf{a}}'(\theta))^H \right| \right. \\ & \quad \left. + \text{tr} \left(\left[\hat{\Sigma}_{\mathbf{x}, -k} + \eta \hat{\mathbf{a}}'(\theta) (\hat{\mathbf{a}}'(\theta))^H \right]^{-1} \hat{\mathbf{R}}_{\mathbf{x}} \right) \right\}. \end{aligned} \quad (50)$$

Denote the objective function in (50) by $\mathcal{L}_k$, the estimate of η_k can be derived from $\partial \mathcal{L}_k / \partial \eta = 0$ as

$$\hat{\eta}_k = \frac{(\hat{\mathbf{a}}'(\theta))^H \hat{\Sigma}_{\mathbf{x}, -k}^{-1} (\hat{\mathbf{R}}_{\mathbf{x}} - \hat{\Sigma}_{\mathbf{x}, -k}) \hat{\Sigma}_{\mathbf{x}, -k}^{-1} \hat{\mathbf{a}}'(\theta)}{\left[(\hat{\mathbf{a}}'(\theta))^H \hat{\Sigma}_{\mathbf{x}, -k}^{-1} \hat{\mathbf{a}}'(\theta) \right]^2}. \quad (51)$$

Substituting (51) into $\partial \mathcal{L}_k / \partial \theta = 0$ yields

$$\begin{aligned} & \text{Re} \left\{ (\hat{\mathbf{a}}'(\theta))^H \hat{\Sigma}_{\mathbf{x}, -k}^{-1} \left[\hat{\mathbf{a}}'(\theta) (\hat{\mathbf{a}}'(\theta))^H \hat{\Sigma}_{\mathbf{x}, -k}^{-1} \hat{\mathbf{R}}_{\mathbf{x}} \right. \right. \\ & \quad \left. \left. - \hat{\mathbf{R}}_{\mathbf{x}} \hat{\Sigma}_{\mathbf{x}, -k}^{-1} \hat{\mathbf{a}}'(\theta) (\hat{\mathbf{a}}'(\theta))^H \right] \hat{\Sigma}_{\mathbf{x}, -k}^{-1} \hat{\mathbf{d}}'(\theta) \right\} = 0, \end{aligned} \quad (52)$$

where $\hat{\mathbf{d}}'(\theta) = \partial \hat{\mathbf{a}}'(\theta) / \partial \theta$.

As $\hat{\mathbf{R}}_{\mathbf{x}}$ and $\hat{\Sigma}_{\mathbf{x}, -k}$ are generally inaccurate in practical applications, ϑ_k should be estimated by checking the distance of the item on the left-hand-side of (52) to 0, instead of testing the equivalence, which results in the DOA estimator given in (20).

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